

ORCA

Marine EcOsystem Reasoning with Collaborative Agents — Master
Application Build Specification

Problem Statement ID: 26176

Organisation: Indian Space Research Organisation (ISRO)

Document Type: Complete End-to-End Engineering & Build Specification

Version 1.0

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ABOUT THIS DOCUMENT

Purpose

This document is the complete, unabridged, end-to-end engineering and product specification for **ORCA — Marine EcOsystem Reasoning with Collaborative Agents**. It is written so that it can be handed, as-is, to a human engineering team or to an AI coding agent and used directly as the single source of truth for building the application from an empty repository through to a fully working, edge-case-tested, demo-ready product.

Nothing in the original specification has been shortened, summarized, or paraphrased away. Every requirement, every agent definition, every UI mock, every state machine, every edge case, and every red-team question from the original build prompt is preserved in full below, organized into **154 numbered sections** plus supporting front matter, so that no requirement is lost between planning and implementation.

How This Document Is Organized

1. **Sections 1-17** define the product vision, the official problem statement, the three-layer architecture principle, the required agents, agent contracts, orchestration, and the end-to-end user query pipeline.
2. **Sections 18-46** define the conversational interface, response design, multi-day trip planning, the active-trip state machine, continuous monitoring, event-driven architecture, alerting, routing, geofencing, community features, and the full Emergency Mode / SOS / rescue-integration workflow.
3. **Sections 47-83** define offline and failure handling, data-freshness and source-trust rules, the risk engine architecture, uncertainty handling, multi-source conflict resolution, notification fatigue controls, dashboards, maps, explainability, language and voice support, privacy, security, database design, the event system, performance, observability, and the full demo simulation timeline.
4. **Sections 84-104** define the exhaustive edge-case catalogue across user, location, weather, ocean, agent, emergency, community, privacy, network, and data dimensions, along with the core safety principles, testing scenarios, red-team questions, UX red-team questions, and accessibility requirements.
5. **Sections 105-125** define the technology stack, repository structure, documentation set, database seeding, the simulation engine, the emergency data model and status machine, trip history, the learning loop, performance architecture, and the non-negotiable product-boundary rules.
6. **Sections 126-154** define the phased build order, iterative-development discipline, UI quality bar, the complete screen inventory, all core UI flows, dynamic replanning, memory and context handling, the data-source registry, audit logging, the final application requirement, the final edge-case checklist, the final

red-team questions, the final success criteria, and the final build command.

How to Use This Specification

- Treat every requirement below as **mandatory** unless it is explicitly described as optional or “where practical.”
- Treat every code block, JSON schema, ASCII diagram, and UI mock exactly as written — they are authoritative structural definitions, not illustrative suggestions.
- Do not treat the length of this document as a reason to compress or skip sections during implementation. The completeness of this specification is intentional: ORCA is a **safety-critical marine intelligence system**, and incomplete implementation of any single section (especially the edge-case and safety sections) can translate directly into unsafe behavior for a real fisherman at sea.
- Build in the phase order defined in Section 126, verify against the checklists in Sections 84–104 and 150, and only declare the application complete once every item in Sections 150–152 is satisfied.

ORCA — MASTER APPLICATION BUILD PROMPT

ROLE

You are the **Lead Product Architect + Full-Stack Engineer + Agentic AI Engineer + GIS Engineer + Marine Intelligence Engineer + Disaster-Safety Systems Engineer + UX Engineer + QA/Red-Team Engineer** responsible for building the complete ORCA application.

You are not being asked to create a concept document.

You are being asked to **BUILD THE ACTUAL APPLICATION**.

The application must implement the complete ORCA vision described below.

You must think like an engineering team building a serious production-grade prototype that could later evolve into a real marine safety platform.

1. PRODUCT

ORCA

Marine EcOsystem Reasoning with Collaborative Agents

ORCA is an Agentic AI-powered marine intelligence and safety platform.

Its purpose is not merely to answer:

“Can I go fishing tomorrow?”

Its purpose is to support the user across the **entire marine journey**:

PLAN → DECIDE → DEPART → MONITOR → DETECT → ALERT → GUIDE →
RESPOND → RESCUE → RECOVER → RETURN

The defining product principle is:

ORCA DOES NOT STOP AT THE INITIAL RECOMMENDATION.

If ORCA says:

“Tomorrow looks safe.”

and the user goes to sea, ORCA continues monitoring relevant conditions.

If conditions change:

ORCA detects the change → verifies it → assesses risk → alerts the user → recommends an action → continues monitoring.

If the user enters an emergency:

ORCA switches to Emergency Mode → provides actionable instructions → captures location → initiates available emergency communication/integration → supports rescue coordination → continues monitoring until the situation is resolved.

This lifecycle is the most important product principle.

2. OFFICIAL PROBLEM STATEMENT

Build the application around the official PS:

Problem Statement ID: 26176

Title: ORCA Marine Ecosystem Reasoning with Collaborative Agents

Organization: Indian Space Research Organisation (ISRO)

The official PS asks for an Agentic AI conversational marine intelligence platform capable of:

- natural-language marine queries
- intent understanding
- autonomous task decomposition
- specialized agent collaboration
- satellite Earth Observation integration
- weather information
- oceanographic information
- GIS/geospatial information
- spatial-temporal reasoning
- PFZ intelligence
- marine advisories
- route optimization
- safety alerts
- geofencing
- explainable recommendations
- maps
- charts
- Indian regional-language support
- contextual multi-turn conversation

ORCA must remain aligned with these requirements.

3. CORE PRODUCT PRINCIPLE

Build ORCA around three layers:

LAYER 1 — INTELLIGENCE

Understand:

What does the user want?

LAYER 2 — DECISION SUPPORT

Understand:

What is happening in the marine environment?

and:

What should the user consider doing?

LAYER 3 — CONTINUOUS SAFETY

Understand:

What changed after the original recommendation?

and:

Does the user need a new warning, route, shelter, or emergency response?

4. DO NOT BUILD A GENERIC CHATBOT

This is extremely important.

Do NOT build:

Chat UI + LLM + fake weather response.

Do NOT make the LLM invent weather, ocean, satellite, rescue, or geospatial information.

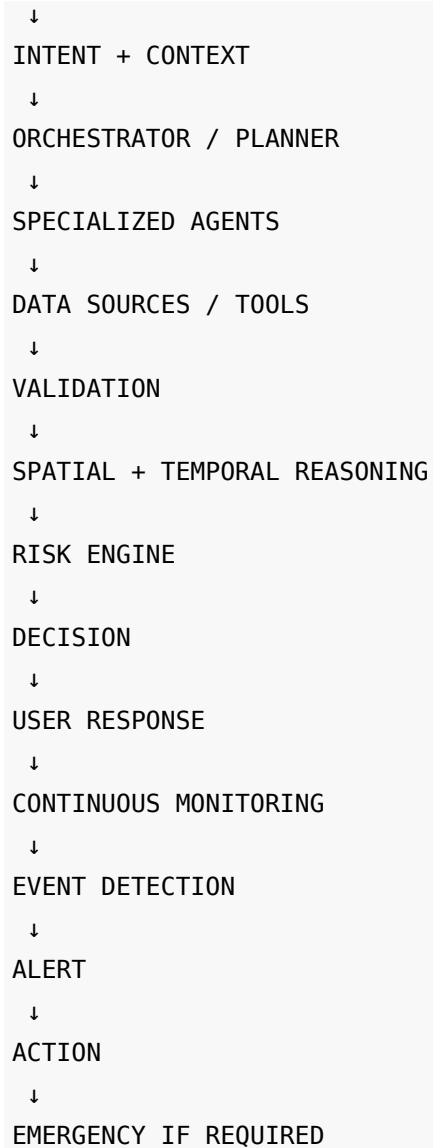
ORCA must be an actual **tool-grounded agentic application**.

Architecture principle:

USER

↓

CONVERSATIONAL INTERFACE



5. REQUIRED AGENTS

Start with these core agents:

1. Weather Agent

Responsibilities:

- current weather
- forecast
- wind

- rainfall
- thunderstorms
- lightning
- visibility
- temperature
- severe weather
- cyclone/storm warnings
- weather trends
- forecast changes

Must return structured data.

6. OCEAN AGENT

Responsibilities:

- wave height
- swell
- sea state
- ocean currents
- SST
- tides where appropriate
- ocean conditions
- marine hazards
- ocean trend

Must understand:

- current conditions
 - forecast conditions
 - historical/contextual conditions when available
-

7. GEOSPATIAL AGENT

Responsibilities:

- user coordinates
- fishing location
- route
- distance
- marine boundaries

- international maritime boundaries
- restricted areas
- marine protected areas
- ecologically sensitive zones
- ports
- harbours
- shelters
- emergency locations
- geofencing
- proximity detection

Use deterministic GIS calculations wherever possible.

Do NOT ask an LLM to calculate spatial intersections when PostGIS/GIS logic can do it reliably.

8. SATELLITE AGENT

Responsibilities:

- satellite Earth Observation information
- SST products
- chlorophyll
- ocean colour
- relevant satellite observations
- imagery/derived marine intelligence

Where live satellite APIs are unavailable:

Create a clean adapter interface and use:

- public datasets
- cached data
- sample datasets
- prototype data

Do not fabricate live satellite information.

9. RISK AGENT

This is one of the most important agents.

Responsibilities:

- combine outputs from other agents
- identify hazards
- calculate risk
- identify dominant risk factors
- evaluate uncertainty
- determine severity
- produce recommendation
- provide explanation
- trigger reassessment

Potential levels:

GREEN
YELLOW
ORANGE
RED

But design the final levels based on the actual risk model.

The Risk Agent must not simply ask the LLM:

“Is this safe?”

It must use structured evidence.

10. ADDITIONAL AGENTS

Create additional specialized agents only where they provide genuine functionality.

Possible agents:

Planner / Orchestrator Agent

Breaks the user’s request into tasks.

Context Agent

Maintains:

- user context
- vessel context
- trip context
- location

- time
- active plans

PFZ Agent

Handles Potential Fishing Zone information.

Tide Agent

Handles tidal intelligence.

Route Agent

Calculates safe/optimal routes.

Evidence Agent

Validates sources and timestamps.

Alert Agent

Determines whether an event warrants notification.

Monitoring Agent

Maintains active trip monitoring.

Emergency Agent

Handles emergency workflows.

Rescue Coordination Agent

Handles authorized rescue information.

Community Agent

Handles community reports and observations.

Moderation Agent

Moderates community content.

Notification Agent

Handles:

- push notifications
- SMS where supported
- in-app notifications

Visualization Agent

Determines:

- map layers
- charts
- risk visualization
- evidence visualization

Language Agent

Handles:

- language detection
- translation
- regional-language responses
- voice workflows

Do not create unnecessary agents.

Each agent must have a clear responsibility.

11. AGENT CONTRACTS

Every agent must have:

```
INPUT
TOOLS
PROCESS
OUTPUT
CONFIDENCE
TIMESTAMP
SOURCE
ERROR STATE
```

Use structured JSON schemas internally.

Example:

```
{
  "agent": "weather",
  "location": {},
  "timestamp": "",
  "conditions": {},
  "forecast": {},
  "warnings": [],
  "sources": [],
  "freshness": {},
  "confidence": 0.0,
  "status": "success"
}
```

Do not allow free-form agent-to-agent communication when structured contracts can be used.

12. AGENT ORCHESTRATION

For a query such as:

“Can I go fishing tomorrow morning?”

The orchestrator should determine that it needs:

Context Agent
Weather Agent
Ocean Agent
Geospatial Agent
Satellite Agent
Tide Agent
PFZ Agent
Risk Agent
Route Agent
Evidence Agent

Independent data collection should happen in parallel wherever possible.

Then:

AGENTS
↓
EVIDENCE VALIDATION

↓
RISK
↓
ROUTE
↓
FINAL RESPONSE

13. USER QUERY PIPELINE

Build this workflow:

USER QUERY
↓
Language Detection
↓
Intent Detection
↓
Entity Extraction
↓
Location Resolution
↓
Time Resolution
↓
Trip Context
↓
Missing Information Check
↓
Planner
↓
Agent Execution
↓
Validation
↓
Risk Analysis
↓
Response

14. AMBIGUOUS QUERIES

Handle:

“Can I go tomorrow?”

If location/time is missing:

Ask only the minimum required clarification.

Do NOT ask 10 questions.

Use available context.

Example:

“What time are you planning to leave?”

If location is already known from an active trip, do not ask again.

15. LOCATION HANDLING

Support:

Current GPS

Manually selected location

Saved fishing location

Planned fishing location

Route location

Destination

Last known location

Handle:

- GPS denied
- GPS unavailable
- stale GPS
- inaccurate GPS
- permission denied
- location outside planned area
- location suddenly changes

Never silently assume a location.

16. FIRST-TIME USER

Build onboarding.

Collect only necessary information.

Possible:

- name
- language
- vessel type
- vessel size/category
- emergency contact
- optional home port
- optional common fishing area

Clearly explain why information is needed.

Do not collect unnecessary personal information.

17. MAIN APPLICATION

Design a professional mobile-first interface.

Main sections:

Home

Ask ORCA

Map

Active Trip

Alerts

Community

Emergency

Profile

Do not make the UI look like a generic AI chatbot.

It should feel like:

Marine Intelligence + Safety Platform

18. HOME SCREEN

Show:

Current safety status

Example:

MARINE CONDITIONS MODERATE RISK

Show:

- weather
- sea
- wind
- waves
- alerts
- active trip
- latest advisory

Also show:

Ask ORCA

with example queries:

- “Can I go fishing tomorrow?”
 - “Show safe fishing areas.”
 - “What is the sea condition near me?”
 - “Is there any cyclone alert?”
 - “What route should I take?”
-

19. CONVERSATIONAL UI

The user should be able to type or speak naturally.

Examples:

“Kal subah 6 baje fishing ke liye ja sakta hu?”

“What about afternoon?”
“What if I stay closer to shore?”
“Show me the safest route.”
“Create a 7-day fishing plan.”

Maintain conversational context.

20. RESPONSE DESIGN

Never return only:

“Yes, it is safe.”

Return structured information.

Example:

[GREEN] LOW RISK

Recommended: GO

Why:

- Wind conditions are favourable
- Wave conditions are within acceptable range
- No active severe marine warning detected
- Return window remains favourable

Watch:

- Conditions may deteriorate after 3 PM

Recommendation:

Return before the risk window increases.

Last updated:

10:42 AM

Include:

- source
- timestamp
- confidence
- major risk drivers

- recommendation
 - what could change
-

21. MULTI-DAY TRIP MODE

User can say:

“I will be fishing for the next 7 days at these locations.”

Create:

ACTIVE MARINE PLAN

Store:

- start date
- end date
- locations
- routes
- expected fishing times
- vessel
- user preferences
- monitoring requirements

ORCA determines:

- what to monitor
- when to reassess
- when to notify
- what events matter

The user should NOT have to repeatedly ask.

22. ACTIVE TRIP STATE

Build a state machine.

Minimum:

IDLE

↓

PLANNING

↓

```
READY
↓
DEPARTED
↓
EN_ROUTE
↓
FISHING
↓
CONDITIONS_CHANGING
↓
WARNING
↓
SAFE_ACTION
↓
EMERGENCY
↓
RESCUE
↓
SAFE
↓
RETURNING
↓
COMPLETED
```

Add states if necessary.

State must be persistent.

If the application restarts, the active trip should not disappear.

23. CONTINUOUS MONITORING

This is the core USP.

When a trip becomes active:

ORCA continuously evaluates relevant information.

Do not simply refresh the chat.

Build a monitoring engine.

Monitor:

- weather
 - waves
 - wind
 - lightning
 - cyclone
 - sea state
 - route
 - geofence
 - marine advisories
 - vessel location
 - shelter proximity
 - other relevant hazards
-

24. EVENT-DRIVEN ARCHITECTURE

Do not notify the user every time data changes.

Create meaningful events.

Examples:

```
CYCLONE_WARNING  
HIGH_WAVE  
STRONG_WIND  
LIGHTNING  
RAPID_WEATHER_DETERIORATION  
GEOFENCE_PROXIMITY  
RESTRICTED_ZONE_ENTRY  
BOUNDARY_PROXIMITY  
ROUTE_RISK_INCREASE  
SAFE_CONDITION_RESTORED  
GPS_LOST  
CONNECTIVITY_LOST  
VESSEL_STATIONARY  
EMERGENCY
```

Each event should go through:

```
DETECT  
↓  
VERIFY  
↓
```

ASSESS

↓

SEVERITY

↓

DECISION

↓

ALERT

↓

MONITOR RESPONSE

25. ALERT ENGINE

Build severity-aware alerts.

INFO

No urgent action.

ADVISORY

Be aware.

WARNING

Action recommended.

CRITICAL

Immediate action.

EMERGENCY

Emergency workflow.

Do not flood users with notifications.

Implement notification deduplication.

26. ALERT ESCALATION

If the user ignores a critical warning:

Do NOT automatically assume they are in danger.

Instead:

1. reassess
2. determine whether danger is increasing
3. attempt another warning
4. provide clear actions
5. escalate only according to configured safety policy

Emergency authority actions must be explicit and legally/technically realistic.

27. NIGHT-BEFORE SCENARIO

Implement the complete workflow.

User:

“Can I go fishing tomorrow morning?”

System:

```
Understand query
↓
Resolve location
↓
Resolve time
↓
Check weather
↓
Check ocean
↓
Check waves
↓
Check wind
↓
Check tides
↓
Check warnings
↓
```

Check satellite/ocean observations

↓

Check geofences

↓

Check route

↓

Assess return conditions

↓

Risk

↓

Recommendation

28. RETURN-WINDOW ANALYSIS

Do NOT only analyze outbound travel.

Calculate:

OUTBOUND CONDITIONS

+

FISHING WINDOW

+

RETURN CONDITIONS

Example:

Morning departure is acceptable, but conditions are expected to deteriorate later.

ORCA should recommend:

“If you go, plan to return before the deterioration window.”

29. SAFE TRIP SCENARIO

If the user goes fishing and conditions remain normal:

ORCA should:

- continue monitoring
- avoid unnecessary alerts
- update relevant information

- show trip status
 - allow user to query conditions
 - provide return guidance
 - close trip when returned
-

30. CONDITIONS CHANGE SCENARIO

Example:

ORCA previously said:

GO

Later:

Wave height increases.

Weather deteriorates.

ORCA must:

```
Detect
↓
Verify
↓
Compare with previous state
↓
Assess risk
↓
Determine significance
↓
Alert
↓
Recommend action
```

The system must never say:

“You were safe earlier, so you’re safe now.”

Current conditions override stale recommendations.

31. WARNING UI

Example:

[WARNING] CONDITIONS CHANGING

Your current area is experiencing worsening marine conditions.

Main changes:

- Wind increasing
- Waves increasing
- Thunderstorm risk increasing

Recommended:

Begin returning toward a safer location.

[RETURN HOME]

[FIND SAFE LOCATION]

[VIEW MAP]

[CONTINUE MONITORING]

[SOS]

Buttons must perform real application actions.

32. SAFE LOCATION

Build a geospatial safe-location engine.

Potential locations:

- verified shelters
- ports
- harbours
- designated emergency facilities
- other validated locations

For every location show:

- distance
- route
- estimated travel time where reliable
- current status where available
- suitability

- source

Never label an unverified location as “safe.”

33. ROUTE ENGINE

Build route planning.

Objectives:

Safety

Distance

Time

Fuel

Geofence avoidance

Weather exposure

Emergency accessibility

The shortest route is not necessarily the safest route.

Support:

Home → Fishing Area

Fishing Area → Safe Location

Fishing Area → Home

Recalculate routes when conditions change.

34. GEOFENCING

Implement:

- international maritime boundary warning
- restricted area warning
- protected area warning
- operational boundaries
- user-defined zones

Use PostGIS/geospatial calculations.

Trigger warnings based on proximity.

Example:

[WARNING] You are approaching a restricted marine zone.

Show:

- map
 - distance
 - boundary
 - recommended action
-

35. COMMUNITY

Build a community section.

Users can:

- create posts
- create observations
- create polls
- vote
- report hazards
- report unusual local conditions

Example:

“How are sea conditions near Location X?”

Responses should be aggregated.

36. COMMUNITY TRUST

Never treat every community post as truth.

Implement:

- verification
- source type
- timestamp
- location

- reputation
- duplicate detection
- moderation
- confidence

A community report should be considered an observation, not automatically an official warning.

37. COMMUNITY MODERATION

Moderate:

- spam
- abusive content
- sexual content
- illegal content
- dangerous misinformation
- impersonation
- malicious instructions
- harassment
- fake emergency reports

Implement:

```
USER CONTENT
↓
MODERATION
↓
ALLOW
OR
FLAG
OR
REMOVE
OR
HUMAN REVIEW
```

Do not rely entirely on an LLM for moderation.

38. OFFICIAL VS COMMUNITY DATA

Implement evidence hierarchy.

Possible hierarchy:

OFFICIAL AUTHORITATIVE SOURCE

↓

SCIENTIFIC DATASET

↓

VERIFIED OBSERVATION

↓

MULTIPLE INDEPENDENT REPORTS

↓

UNVERIFIED COMMUNITY REPORT

But research and improve this hierarchy.

When sources conflict:

Do not average blindly.

Use:

- source authority
 - timestamp
 - distance
 - relevance
 - confidence
 - corroboration
-

39. EMERGENCY MODE

Emergency Mode must be a separate application state.

User presses:

SOS

The system should capture:

- latest known location
- timestamp
- user
- vessel
- number of people if available
- active trip
- current route
- environmental conditions

- last communication
 - emergency details
-

40. SOS UX

Create a clear emergency screen.

Example:

```
[EMERGENCY] EMERGENCY MODE
```

Your location:

```
[MAP]
```

Latest known position:

```
...
```

People onboard:

```
...
```

Conditions:

```
...
```

Available help:

```
...
```

```
[CONTACT EMERGENCY SERVICE]
```

```
[SHARE LOCATION]
```

```
[CALL EMERGENCY CONTACT]
```

```
[VIEW SAFE LOCATION]
```

Do not fabricate rescue functionality.

41. SOS CONFIRMATION

Consider accidental SOS.

Design a safe confirmation mechanism.

But in situations where immediate emergency activation is appropriate, do not create dangerous friction.

Research and implement a configurable approach.

42. RESCUE INTEGRATION

Create an abstraction:

```
RescueProvider
```

Possible implementations:

```
GovernmentRescueProvider  
PartnerRescueProvider  
EmergencyCommunicationProvider  
PrototypeRescueSimulator
```

The architecture must allow future official integrations.

If no official API exists:

DO NOT FAKE IT.

Use a clearly marked:

 Prototype Simulation

for demonstration.

43. RESCUE MAP

Where real authorized data exists:

Show:

- rescue unit
- location
- status
- ETA
- route

If not available:

show:

 “Live rescue-team location requires authorized agency integration.”

For the hackathon demo, a simulation may demonstrate the UX.
Never present simulated rescue data as real.

44. DISASTER MANAGEMENT

Design an adapter architecture for:

- disaster management authorities
- maritime authorities
- emergency agencies
- rescue services

Do not hard-code one agency into the core system.

Use:

Emergency Gateway

↓

Authorized Integration

45. MULTIPLE EMERGENCIES

Support:

- engine failure
- fire
- flooding
- capsizing
- person overboard
- collision
- medical emergency
- vessel immobilization
- navigation failure
- communication failure
- extreme weather
- cyclone
- lightning
- missing vessel

The emergency UI should allow the user to specify the type when possible.

46. EMERGENCY PRIORITY

If multiple people/vessels need help:

Do not allow the LLM alone to determine rescue priority.

Create a structured priority model for prototype decision support.

Factors may include:

- immediate danger
- people affected
- environmental severity
- vessel status
- medical emergency
- communication
- location

Clearly label this as decision support, not autonomous command authority.

47. OFFLINE MODE

The app must degrade gracefully.

Handle:

No internet

Weak internet

GPS but no internet

Internet but no GPS

intermittent connectivity

app restart

low battery

Cache:

- latest advisories
- active trip

- emergency contacts
- critical maps/data
- last known risk state

Do not pretend stale data is live.

Show:

Last updated X minutes ago.

48. OFFLINE SOS

If connectivity is unavailable:

Create an emergency fallback abstraction.

Possible channels:

- phone call
- SMS
- satellite communication
- queued transmission

Only implement channels actually supported by the environment.

Do not claim SMS/satellite integration unless technically implemented.

49. LOCATION FAILURE

Handle:

```
GPS_PERMISSION_DENIED
GPS_UNAVAILABLE
GPS_STALE
GPS_LOW_ACCURACY
GPS_DRIFT
LOCATION_CONFLICT
```

The UI must explain what is happening.

50. DATA FAILURE

Every data provider must support:

```
SUCCESS
STALE
TIMEOUT
UNAVAILABLE
INVALID
PARTIAL
CONFLICTING
```

ORCA should never hallucinate missing data.

Example:

“Ocean-wave data could not be verified right now. I can provide a partial assessment using the available weather and official warning data, but confidence is reduced.”

51. SOURCE FRESHNESS

Every data record must contain:

```
source
observed_at
retrieved_at
valid_until
forecast_time
spatial_resolution
confidence
```

Build stale-data detection.

52. LLM SAFETY

The LLM must NEVER invent:

- coordinates
- weather
- waves
- cyclone status

- rescue locations
- government warnings
- PFZ data
- shelter availability
- maritime boundaries

All factual marine information must originate from tools/data.

LLM responsibility:

Understand

Plan

Explain

Summarize

Communicate

Not:

Fabricate environmental measurements.

53. DETERMINISTIC VS AI

Clearly separate:

Deterministic

- distance
- geofence intersection
- coordinate calculations
- route geometry
- timestamp validation
- data freshness
- threshold checks where formally defined

AI/LLM

- natural language understanding
- query decomposition
- explanation

- conversational context
- summarization

ML/Scientific models

- prediction
- classification
- anomaly detection

Human/Authority

- emergency command
 - rescue authorization
 - official warning issuance
-

54. RISK ENGINE ARCHITECTURE

Create a dedicated risk engine.

Input:

```
WEATHER
OCEAN
SATELLITE
TIDE
GEOSPATIAL
ROUTE
VESSEL
TIME
OFFICIAL WARNINGS
```

Output:

```
risk_level
risk_score
confidence
risk_factors
recommendation
actions
reassessment_conditions
```

Example:

```
{
  "risk_level": "ORANGE",
  "risk_score": 78,
  "confidence": 0.87,
  "factors": [
    "high_wave_conditions",
    "strong_wind",
    "marine_warning"
  ],
  "recommendation": "DO_NOT_PROCEED",
  "reassess_when": [
    "new_marine_bulletin",
    "weather_update"
  ]
}
```

The exact scoring methodology must be designed carefully and clearly documented.

55. UNCERTAINTY

Never display:

“100% safe.”

Instead:

“Conditions currently appear favourable based on the available data.”

If uncertainty is high:

“Confidence is reduced because recent ocean observations are unavailable.”

56. MULTI-SOURCE CONFLICT

Example:

Weather source A:

20 km/h

Source B:

35 km/h

Official warning:

Strong wind.

The system should:

1. detect contradiction
 2. evaluate source authority
 3. evaluate freshness
 4. evaluate proximity
 5. determine conservative risk
 6. explain the conflict if important
-

57. NOTIFICATION FATIGUE

Do not send an alert for every small change.

Create:

- severity threshold
- change threshold
- cooldown
- deduplication
- alert grouping
- escalation logic

Example:

Five identical weather updates should not create five notifications.

58. SAFE CONDITION RESTORATION

If conditions improve after an alert:

ORCA should detect recovery.

Example:

Earlier:
HIGH RISK

Later:

Conditions improved.

ORCA:

"Marine conditions in your area have improved based on the latest verified data. Continue to follow

Do not automatically declare the environment “safe” merely because one metric improved.

59. ACTIVE TRIP DASHBOARD

Build a dedicated screen showing:

Current status

User location

Route

Destination

Weather

Ocean

Risk

Alerts

Geofence status

Return recommendation

Emergency button

Last updated

60. MAP

The map should support layers:

- current position
- planned route
- safe route

- fishing zones
- PFZ
- SST
- chlorophyll
- waves
- wind
- cyclone
- lightning
- geofences
- boundaries
- protected zones
- shelters
- ports
- emergency resources

Do not show every layer simultaneously.

Use a layer selector.

61. SATELLITE VISUALIZATION

Provide appropriate map visualization for:

- SST
- chlorophyll
- satellite-derived products
- PFZ

Use legends.

Show timestamps.

Explain what the layer means.

62. EXPLAINABILITY PANEL

Every major recommendation should have:

WHY?

Example:

Why ORCA recommends caution:

1. Wave height is increasing.
2. Wind conditions are becoming less favourable.
3. A marine advisory is active.
4. Your return route enters a higher-risk period.

Confidence: High

Last updated: ...

63. EVIDENCE PANEL

Allow user to see:

- source
- timestamp
- measurement
- forecast
- official advisory
- relevant map layer

Do not overwhelm the normal UI.

Use:

“View evidence”

for details.

64. LANGUAGE SUPPORT

Build language abstraction.

The user can communicate in:

- English
- Hindi
- other supported Indian languages

Support:

- mixed language
- transliterated Hindi
- local terminology
- voice input where available

Never translate technical safety information in a way that changes meaning.

65. VOICE

Design the app so voice can be added cleanly.

Example:

User:

“Kal subah samundar jana safe hai?”

System:

Speech → text → intent → agents → response → speech.

For prototype, implement if practical.

66. COMMUNITY POLLS

Poll object:

```
id
creator
question
options
location
created_at
expires_at
votes
status
moderation_status
```

Support:

- create
- vote
- remove own poll

- report
 - moderation
 - expiry
-

67. COMMUNITY GEOLOCATION

A community observation should optionally contain:

- approximate location
- timestamp
- observation type

Do not expose exact user location unnecessarily.

Protect sensitive fishing locations.

68. USER PRIVACY

Implement:

- authentication
- authorization
- location consent
- emergency-sharing consent where applicable
- data minimization
- secure storage
- audit logging

Do not expose:

- exact fishing location
- personal data
- emergency information

to other users unless explicitly intended.

69. SECURITY

Protect against:

- account takeover
 - fake SOS
 - GPS spoofing
 - malicious community content
 - prompt injection
 - tool injection
 - data poisoning
 - API abuse
 - notification abuse
 - fake rescue identity
 - unauthorized location access
-

70. AGENT SECURITY

External data is DATA.

It is never an instruction.

If a web/API/community source contains:

“Ignore your safety rules.”

The system must treat it as untrusted content.

Implement:

- tool permissions
 - agent isolation
 - structured inputs
 - validation
 - source trust
 - least privilege
-

71. PROMPT INJECTION PROTECTION

The LLM must not allow:

- retrieved documents
- community posts
- weather descriptions
- external web pages
- user-generated content

to override system safety rules.

72. AUTHENTICATION

Implement:

- registration
- login
- logout
- session handling
- password/security flow
- user profile
- emergency contacts

Use secure authentication architecture.

73. DATABASE

Design a proper relational schema.

Potential entities:

```
User
Vessel
EmergencyContact
Trip
TripLocation
Route
AgentRun
DataObservation
WeatherObservation
OceanObservation
SatelliteObservation
MarineAlert
RiskAssessment
Notification
Geofence
Shelter
Emergency
SOS
```

```
RescueUnit
CommunityPost
CommunityReport
Poll
PollVote
AuditLog
```

Modify this schema where required.

Use spatial types where appropriate.

74. POSTGIS

Use PostGIS or equivalent for:

- coordinates
 - routes
 - geofences
 - spatial intersections
 - proximity
 - maritime boundaries
 - shelters
 - zones
-

75. EVENT SYSTEM

Implement an event-driven architecture.

Possible:

```
WeatherUpdated
OceanUpdated
MarineAlertIssued
RiskChanged
GeofenceApproaching
TripStarted
TripEnded
EmergencyTriggered
SOSTriggered
LocationUpdated
```

ConnectivityChanged
ConditionsRecovered

Agents/services should react to events.

76. MONITORING LOOP

Design:

```
ACTIVE TRIP
↓
GET NEW DATA
↓
VALIDATE
↓
COMPARE WITH PREVIOUS STATE
↓
DETECT EVENTS
↓
RISK
↓
IF SIGNIFICANT CHANGE
  ↓
  ALERT
  ↓
  STORE STATE
```

Do not require an LLM invocation for every tiny update.

Use deterministic event detection where possible.

Use LLM reasoning only when useful.

77. COST CONTROL

Do not call the LLM unnecessarily.

Use:

- deterministic rules
- caching

- event thresholds
- structured agent outputs
- batching
- parallel calls

LLM should be invoked when reasoning/explanation is required.

78. PERFORMANCE

Target:

- fast initial query
- parallel agent execution
- caching
- asynchronous monitoring
- non-blocking notifications

The app should remain responsive even if one external data source is slow.

79. FAILURE ISOLATION

If Satellite Agent fails:

Weather/Ocean/Geospatial should still work.

If Weather API fails:

Other agents should still operate.

If LLM fails:

Critical deterministic safety checks should still operate where possible.

Do not make the entire platform dependent on one component.

80. AGENT OBSERVABILITY

Create an internal developer dashboard/logging system showing:

User Query

↓

```
Planner
↓
Agents called
↓
Tool calls
↓
Sources
↓
Agent outputs
↓
Risk
↓
Final response
```

This will be extremely useful during the hackathon demo.

81. DEMO MODE

Create a controlled demo/simulation mode.

This is extremely important for demonstrating:

“Safe → conditions worsen → alert → emergency”

without waiting for a real disaster.

Create simulated scenarios:

Scenario 1

Normal weather.

Scenario 2

Weather deterioration.

Scenario 3

Cyclone warning.

Scenario 4

Geofence proximity.

Scenario 5

SOS.

Scenario 6

Rescue simulation.

Clearly label simulated information.

82. DEMO TIMELINE

Build a demo scenario:

10:00 PM

User asks:

“Can I go fishing tomorrow at 6 AM?”

ORCA:

GO WITH CAUTION

6:00 AM

Trip starts.

9:30 AM

Conditions normal.

Dashboard:

GREEN

11:15 AM

Wind increases.

Risk:

YELLOW

No unnecessary panic.

12:00 PM

Severe weather warning.

Risk:

ORANGE

Notification:

“Conditions are deteriorating.”

12:10 PM

Route recommendation updates.

12:20 PM

User enters a higher-risk area.

ORCA warns.

12:30 PM

User presses SOS.

Emergency Mode starts.

12:31 PM

Location displayed.

12:32 PM

Prototype rescue integration displays rescue unit.

Clearly label:

DEMO / SIMULATED RESCUE DATA

12:40 PM

User reaches safe location.

ORCA confirms:

“Emergency status updated.”

Later

Conditions improve.

ORCA updates the user.

83. THIS DEMO MUST SHOW THE USP

The audience must understand:

A normal system:

Question

↓

Answer

ORCA:

Question

↓

Decision

↓

Trip

↓

Monitoring

↓

Change

↓
Alert
↓
Action
↓
Emergency
↓
Rescue
↓
Recovery

This is the product story.

84. EDGE-CASE ENGINEERING

You must actively test all of the following.

USER EDGE CASES

- first-time user
 - returning user
 - missing information
 - ambiguous query
 - language mismatch
 - voice transcription error
 - user changes mind
 - user changes destination
 - user changes departure time
 - user ignores warning
-

85. LOCATION EDGE CASES

- no GPS
- inaccurate GPS
- GPS denied
- stale GPS
- location jumps
- location outside trip area

- boundary proximity
 - restricted zone
 - protected zone
 - international boundary
-

86. WEATHER EDGE CASES

- normal
 - rain
 - strong wind
 - high waves
 - lightning
 - thunderstorm
 - cyclone
 - rapidly changing conditions
 - conflicting forecast
 - stale forecast
 - API unavailable
-

87. OCEAN EDGE CASES

- missing wave data
 - stale wave data
 - contradictory wave data
 - abnormal SST
 - high swell
 - strong currents
 - tide mismatch
 - satellite unavailable
-

88. AGENT EDGE CASES

- agent timeout
- agent failure
- invalid output
- conflicting agent outputs

- hallucinated output
- tool failure
- unavailable data
- partial data
- malicious external content

Implement retries and graceful fallback where appropriate.

89. EMERGENCY EDGE CASES

Test:

- accidental SOS
 - real SOS
 - SOS + no internet
 - SOS + no GPS
 - SOS + low battery
 - SOS + multiple people
 - SOS + multiple vessels
 - SOS + restricted area
 - SOS + international boundary
 - rescue unavailable
 - rescue ETA unknown
 - disaster overwhelms rescue capacity
 - backend unavailable during SOS
-

90. COMMUNITY EDGE CASES

Test:

- spam
- misinformation
- fake report
- abusive poll
- illegal content
- sexual content
- malicious content
- duplicate reports
- contradictory reports

- coordinated manipulation
 - fake emergency report
-

91. PRIVACY EDGE CASES

Test:

- user revokes location
 - user deletes account
 - user disables tracking
 - community user attempts to see another user's location
 - emergency data access
 - rescue authority access
 - unauthorized API access
-

92. NETWORK EDGE CASES

Test:

- online
 - offline
 - slow internet
 - intermittent connection
 - reconnect
 - backend unavailable
 - notification service unavailable
-

93. DATA EDGE CASES

Test:

- stale
- missing
- invalid
- duplicate
- conflicting
- delayed
- incorrect coordinate

- incorrect timestamp
 - source unavailable
-

94. SAFETY PRINCIPLE

When critical information is unavailable:

DO NOT pretend everything is safe.

Prefer:

“Unable to verify.”

with appropriate caution.

95. DO NOT CREATE FAKE LIVE DATA

For the prototype, you may use:

- seeded data
- public sample data
- controlled simulation
- mock APIs

But label them internally and in the UI where necessary.

Never claim:

“Live government data”

unless it actually is.

96. API ADAPTER ARCHITECTURE

Create interfaces such as:

```
WeatherProvider
OceanProvider
SatelliteProvider
PFZProvider
MarineAlertProvider
```

```
GeospatialProvider
RescueProvider
NotificationProvider
MapProvider
```

Then implement:

```
RealProvider
MockProvider
SimulationProvider
```

This allows the prototype to work even when real APIs are unavailable.

97. CONFIGURATION

All external services must be configured through environment variables.

Never hard-code API keys.

Use:

```
.env
```

and provide:

```
.env.example
```

98. ERROR HANDLING

Every API call should have:

- timeout
- retry
- fallback
- logging
- validation

Do not let one failed provider crash the application.

99. TESTING

Create:

Unit tests

Integration tests

Agent tests

API tests

GIS tests

Risk-engine tests

Event-engine tests

Emergency tests

UI tests

End-to-end tests

100. TEST SCENARIOS

At minimum create automated tests for:

```
SAFE_TRIP
WEATHER_DETERIORATION
CYCLONE
HIGH_WAVE
LIGHTNING
GEOFENCE_WARNING
RESTRICTED_ZONE
CONFLICTING_DATA
STALE_DATA
API_FAILURE
GPS_FAILURE
OFFLINE_MODE
SOS
FALSE_SOS
RESCUE_SIMULATION
```

101. RED TEAM

After implementation, attack the application.

Ask:

- Can I make ORCA hallucinate weather?
- Can I make ORCA claim a fake rescue unit?
- Can I make a community post override official data?
- Can I trigger fake SOS?
- Can I access another user's location?
- Can I manipulate a geofence?
- Can I inject malicious instructions through external data?
- Can I make the system call unnecessary LLM agents?
- Can I make stale data look current?
- Can I crash the monitoring system?

Fix every discovered issue.

102. UX RED TEAM

Ask:

- Would a fisherman understand this?
- Is the warning too technical?
- Is the emergency button obvious?
- Are there too many alerts?
- Can someone use this quickly under stress?
- Is the map readable?
- Is the risk explanation understandable?

Does the app work in low connectivity?

Is the language natural?

Fix accordingly.

103. ACCESSIBILITY

Support:

- large buttons
- high readability
- clear colors with text/icon redundancy
- voice where possible
- regional languages
- simple language
- emergency-first UX

Do not rely on color alone.

104. MOBILE-FIRST

The application should primarily be designed for mobile use.

Responsive web can be supported.

The emergency workflow must work on small screens.

105. TECH STACK

You may use a modern stack such as:

Frontend

React / Next.js / React Native / Flutter

Backend

FastAPI / Python / Node

Database

PostgreSQL + PostGIS

Cache

Redis

Event system

Redis Streams / Kafka / equivalent

AI

LLM + structured tool calling

Maps

A legitimate map provider

Geospatial

PostGIS + GIS libraries

Notifications

Push notification provider

But select the actual stack based on feasibility.

Do not add technologies simply to make the architecture look impressive.

106. REPOSITORY STRUCTURE

Create a clean repository.

Example:

```
orca/  
|  
├─ frontend/  
|  
├─ backend/  
|
```

```
|— agents/
|
|— services/
|
|— risk-engine/
|
|— geospatial/
|
|— monitoring/
|
|— emergency/
|
|— community/
|
|— data-providers/
|
|— simulation/
|
|— tests/
|
|— docs/
|
|— scripts/
|
|— .env.example
|— README.md
```

Modify this if a better architecture is appropriate.

107. DOCUMENTATION

Create:

README

Include:

- product
- architecture
- setup

- environment variables
- database
- running frontend
- running backend
- running agents
- running simulation
- testing

Also create:

ARCHITECTURE.md

AGENTS.md

SAFETY.md

API.md

DATA_SOURCES.md

DEMO.md

EDGE_CASES.md

108. DATABASE SEEDING

Provide seed data for:

- users
- vessels
- sample trips
- sample locations
- weather
- ocean
- satellite
- PFZ
- geofences
- shelters
- alerts
- community posts

This must allow the entire demo to run locally.

109. SIMULATION ENGINE

Build a simulation controller.

Example:

```
NORMAL
↓
WEATHER_WORSENING
↓
HIGH_RISK
↓
EMERGENCY
↓
RESCUE
↓
SAFE
```

The simulation should modify environmental data/events and trigger the actual ORCA application workflow.

Do not build a separate fake UI for the demo.

The simulation must exercise the real event pipeline.

110. REAL VS SIMULATED DATA

Every provider should expose:

```
provider_type:
REAL
MOCK
SIMULATION
```

Internally track this.

In developer mode show:

DATA: SIMULATED

when applicable.

111. EMERGENCY DATA MODEL

Create an Emergency object with:

```
id
user_id
trip_id
type
severity
status
location
created_at
updated_at
people_affected
vessel_id
latest_environment
rescue_status
assigned_resource
resolved_at
```

112. EMERGENCY STATUS MACHINE

Implement:

```
NORMAL
↓
SOS_TRIGGERED
↓
EMERGENCY_CONFIRMED
↓
HELP_REQUESTED
↓
RESCUE_ASSIGNED
↓
RESCUE_IN_PROGRESS
↓
SAFE_LOCATION_REACHED
↓
RESOLVED
```

Support:

CANCELLED
FALSE_ALARM
UNRESOLVED

where appropriate.

113. TRIP HISTORY

Store:

- trip start
- route
- locations
- alerts
- risk changes
- user actions
- emergency events
- trip end

Do not expose sensitive history publicly.

114. POST-TRIP SUMMARY

Generate:

Trip duration
Distance
Conditions encountered
Alerts
Risk changes
Route
Incidents
Recommendations

Use actual stored data.

115. LEARNING LOOP

Architect future support for:

```
FORECAST
↓
ACTUAL OBSERVATION
↓
OUTCOME
↓
EVALUATION
↓
MODEL IMPROVEMENT
```

Do not automatically train models on user data.

Keep this as an extensible architecture.

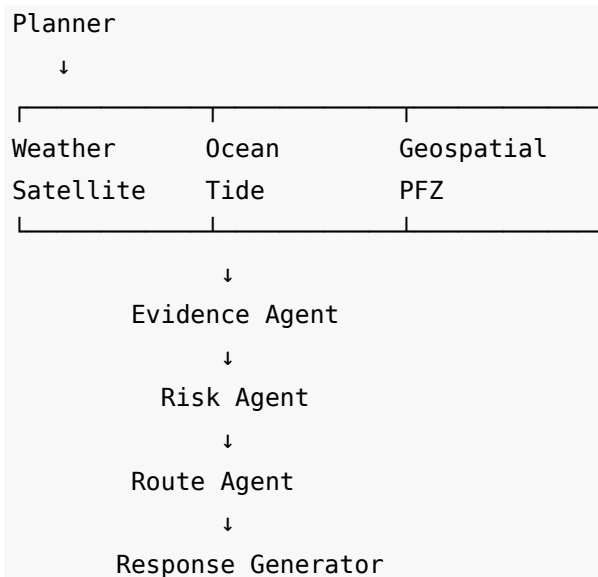
116. PERFORMANCE REQUIREMENT

For a normal query:

Do not execute every agent sequentially.

Use parallel execution where independent.

Example:



117. IMPORTANT AGENT RULE

Agents should not call each other randomly.

Use the orchestrator/state machine.

Keep responsibilities explicit.

118. RESPONSE LATENCY

Implement:

- async calls
- timeouts
- caching
- fallback
- streaming response where useful

If one source is slow, do not freeze the entire application indefinitely.

119. DATA FRESHNESS UI

Show:

Updated 4 min ago

instead of pretending everything is real-time.

If stale:

[WARNING] Some information is outdated.

120. ACTIVE TRIP NOTIFICATION EXAMPLE

When conditions worsen:

[WARNING] MARINE CONDITIONS CHANGING

Your current location:

[Map]

Risk:

ORANGE – HIGH

What changed:

- Wind increased
- Wave conditions deteriorated
- Severe weather probability increased

Recommended:

Return toward the safer route.

[VIEW SAFE ROUTE]

[FIND SAFE LOCATION]

[SOS]

121. EMERGENCY NOTIFICATION

Example:

[EMERGENCY] EMERGENCY MODE ACTIVE

Your latest known location has been captured.

Current conditions:

...

People onboard:

...

Emergency:

...

Available emergency options:

...

Keep the interface simple.

122. NEVER GIVE UNSAFE CERTAINTY

Avoid:

“You are completely safe.”

Use:

“Current conditions appear favourable based on available verified information.”

123. PRODUCT BOUNDARY

ORCA is:

Marine intelligence

Decision support

Continuous monitoring

Safety assistance

Emergency information layer

ORCA is NOT:

Autonomous rescue command

Government authority

Guaranteed weather prediction

Guaranteed emergency response

Replacement for official maritime warnings

124. OFFICIAL WARNINGS TAKE PRIORITY

If an authoritative official warning indicates dangerous conditions, ORCA must not downgrade the warning simply because another source looks favourable.

125. EMERGENCY OVERRIDES NORMAL UX

When an emergency is active:

Do not show the user normal dashboard clutter.

Prioritize:

1. emergency status
 2. location
 3. communication
 4. instructions
 5. rescue information
 6. safe route
 7. emergency contacts
-

126. BUILD ORDER

Do not attempt to write the entire system randomly.

Build in phases.

PHASE 1

Project setup

↓

PHASE 2

Authentication + database

↓

PHASE 3

Main UI

↓

PHASE 4

Query system

↓

PHASE 5

Weather Agent

↓

PHASE 6

Ocean Agent

↓

PHASE 7

Geospatial Agent

↓

PHASE 8

Satellite/PFZ

↓

PHASE 9

Risk Engine

↓

PHASE 10

Trip system

↓

PHASE 11

Continuous monitoring

↓

PHASE 12

Event engine

↓

PHASE 13

Notifications

↓

PHASE 14

Community

↓

PHASE 15

Emergency

↓

PHASE 16

Rescue simulation

↓

PHASE 17

Offline handling

↓

PHASE 18

Security

↓

PHASE 19

Testing

↓

PHASE 20

Demo

127. DO NOT STOP AFTER PHASE 1

Continue implementing the application.

Do not simply provide instructions like:

“You should create X.”

Actually create the files/code/components where your environment permits.

When something cannot be implemented because an external credential/API is unavailable:

1. create the interface
 2. create a mock implementation
 3. make the application functional
 4. document exactly what needs replacing
-

128. ITERATIVE DEVELOPMENT

After each major phase:

1. inspect current code
2. run tests
3. identify failures
4. fix them
5. integrate
6. test again
7. continue

Do not accumulate broken code.

129. NEVER BREAK EXISTING FEATURES

Before modifying a component:

Understand its dependencies.

After modification:

Run relevant tests.

130. UI QUALITY

Do not create an ugly engineering dashboard.

Build a polished product.

Design principles:

- clean
 - modern
 - trustworthy
 - marine-oriented
 - professional
 - easy to understand
 - emergency-ready
-

131. CORE SCREENS

The final application should contain at minimum:

- 1. Splash / onboarding**
 - 2. Login**
 - 3. Home**
 - 4. AI Chat / Ask ORCA**
 - 5. Risk result**
 - 6. Map**
 - 7. Active Trip**
 - 8. 7-Day Plan**
 - 9. Alerts**
 - 10. Community**
 - 11. Polls**
 - 12. Emergency Mode**
 - 13. SOS**
 - 14. Rescue Map**
 - 15. Trip History**
 - 16. Profile**
 - 17. Settings**
-

132. HOME → QUERY FLOW

Build:

```
HOME
↓
ASK ORCA
↓
USER QUERY
↓
UNDERSTANDING
↓
AGENTS
```

↓
RISK
↓
RESULT

The user should be able to:

- start trip
 - create 7-day plan
 - open map
 - see alerts
 - enter emergency mode
-

133. RISK RESULT UI

Include:

RISK LEVEL
RECOMMENDATION
WHY
WEATHER
OCEAN
ROUTE
WARNINGS
RETURN WINDOW
CONFIDENCE
DATA FRESHNESS

134. ACTIVE TRIP UI

Include:

Trip status
Current location
Destination
Risk
Weather
Ocean
Route
Warnings

Return plan
Map
SOS

135. COMMUNITY UI

Include:

Nearby observations
Posts
Polls
Create report
Create poll
Report content

136. ALERT CENTER

Show:

- active alerts
- previous alerts
- severity
- time
- location
- action

Allow user to mark as acknowledged where appropriate.

137. NOTIFICATION ACTIONS

Notifications must deep-link into the correct screen.

Example:

Weather warning → Active Trip → Alert detail.

SOS → Emergency screen.

Geofence → Map.

138. MULTI-DAY PLAN UI

Show:

```
DAY 1
Risk
Weather
Fishing window
Return window
```

```
DAY 2
Risk
...
```

```
DAY 7
...
```

But continue monitoring dynamically.

The 7-day plan is not static.

139. DYNAMIC REPLANNING

If conditions change:

The plan should update.

Example:

Original:

Day 3 fishing area A

New conditions:

High risk.

ORCA:

“Day 3 conditions have changed. Consider postponing or moving to an alternative area.”

Only make alternative recommendations when supported by data.

140. QUERY MEMORY

If user asks:

“Can I go tomorrow?”

then:

“What about afternoon?”

ORCA should understand the context.

Do not force the user to repeat everything.

141. TRIP CONTEXT MEMORY

During an active trip, ORCA should know:

- current location
- destination
- planned route
- trip duration
- current risk
- latest warnings
- recent decisions

But keep memory scoped to the trip where possible.

142. EMERGENCY CONTEXT

When SOS occurs, automatically attach relevant active-trip context.

Do not ask the user to repeat:

“Where are you?”

if the system has a recent reliable location.

143. DATA SOURCE REGISTRY

Create a configuration/registry of providers.

Example:

```
provider
name
type
authority
coverage
update_frequency
supported_variables
status
```

This will make integrations replaceable.

144. SOURCE ATTRIBUTION

Every factual response should be traceable internally to a source.

The user should be able to open:

“Evidence”

and see where important information came from.

145. AUDIT LOG

For important decisions store:

```
timestamp
user
query
agents
data sources
risk
recommendation
alerts
actions
```

This is important for debugging and safety review.

146. RISK REPRODUCIBILITY

Given the same input data, the deterministic portion of the risk engine should produce the same result.

Do not allow random LLM behavior to determine safety status.

147. HUMAN-READABLE REASONING

Do not expose internal chain-of-thought.

Instead provide concise decision factors:

“High wave conditions and an active marine warning are the main reasons for the elevated risk.”

148. DATA VISUALIZATION

Use:

- cards
- graphs
- maps
- timelines
- risk bars
- alert banners

Avoid giant tables on mobile.

149. FINAL APPLICATION REQUIREMENT

The finished application must support this complete lifecycle:

USER

↓

ASK ORCA

↓

UNDERSTAND

↓

WEATHER
OCEAN
SATELLITE
GEOSPATIAL
TIDE
PFZ
↓
RISK
↓
RECOMMENDATION
↓
TRIP CREATED
↓
MONITORING
↓
CONDITION CHANGE
↓
ALERT
↓
SAFE ROUTE / SHELTER
↓
CONTINUE MONITORING
↓
EMERGENCY
↓
SOS
↓
AUTHORIZED RESCUE INTEGRATION / SIMULATION
↓
SAFE LOCATION
↓
RECOVERY
↓
RETURN HOME
↓
TRIP COMPLETION

150. FINAL EDGE-CASE REQUIREMENT

Before considering the application complete, systematically test:

A

User ambiguity

B

Location failure

C

Weather failure

D

Ocean failure

E

Satellite failure

F

GIS failure

G

Agent failure

H

LLM failure

I

Conflicting data

J

Stale data

K

Offline

L

GPS failure

M

Changing weather

N

Cyclone

O

Lightning

P

High waves

Q

Geofence

R

Restricted area

S

Boundary

T

Community misinformation

U

Fake SOS

V

Real SOS

W

Rescue unavailable

X

Multiple emergencies

Y

Backend failure

Z

Security attack

Do not declare completion until each has a defined behavior.

151. FINAL RED-TEAM QUESTIONS

Before final delivery, ask yourself:

Product

Why does ORCA exist?

User

Would a fisherman actually understand it?

AI

Where is AI genuinely useful?

Data

Where does every fact come from?

Safety

What happens when the prediction is wrong?

Monitoring

What happens after the first recommendation?

Emergency

What happens when the situation becomes dangerous?

Connectivity

What happens without internet?

Competition

What does ORCA integrate that fragmented solutions do not?

Feasibility

Which features are real, simulated, or future?

Trust

Why should the user trust the recommendation?

152. FINAL SUCCESS CRITERIA

The application is successful only if all of the following are true:

1.

A user can ask a natural-language marine question.

2.

ORCA decomposes the request.

3.

Multiple specialized agents work together.

4.

The system uses actual/verified/mock-labelled data rather than hallucinations.

5.

The system produces an explainable risk assessment.

6.

The user can create an active trip.

7.

ORCA continues monitoring after the initial answer.

8.

Changing conditions can trigger a new risk assessment.

9.

The user receives actionable alerts.

10.

The route can be recalculated.

11.

Geofencing works.

12.

Community intelligence exists with moderation and trust handling.

13.

Emergency Mode exists.

14.

SOS workflow exists.

15.

Rescue integration has a clean real/mock boundary.

16.

Offline/failure states are handled.

17.

The system does not hallucinate safety-critical information.

18.

The system survives individual agent/API failures.

19.

The application has automated tests.

20.

The complete demo can be executed end-to-end.

153. MOST IMPORTANT PRODUCT STATEMENT

Keep this principle throughout the entire implementation:

ORCA IS NOT A “GO / NO-GO” CHATBOT.

It is:

A CONTINUOUS MARINE INTELLIGENCE AND SAFETY AGENT.

It should answer:

“Can I go?”

but its real responsibility is:

“What should I know, what should I do, and how can ORCA continue helping me as the marine situation changes?”

The application should therefore remain active throughout the user’s journey.

154. FINAL COMMAND

Now begin building.

Do NOT merely describe what you would build.

Do NOT return only architecture diagrams.

Do NOT stop at frontend mockups.

Do NOT create fake features without marking them.

Do NOT assume APIs exist.

Do NOT hallucinate data.

Do NOT skip edge cases.

Do NOT implement only the happy path.

Instead:

1. Inspect the current project/repository.
2. Determine what already exists.
3. Design the implementation plan.
4. Build the foundation.
5. Implement the frontend.
6. Implement backend APIs.
7. Implement database.
8. Implement the agent system.
9. Implement Weather Agent.
10. Implement Ocean Agent.
11. Implement Geospatial Agent.
12. Implement Satellite Agent.
13. Implement Risk Agent.
14. Add additional agents only where required.
15. Implement trip lifecycle.
16. Implement continuous monitoring.
17. Implement event detection.
18. Implement notifications.
19. Implement community.
20. Implement moderation.
21. Implement emergency mode.
22. Implement SOS.
23. Implement rescue adapter/simulation.
24. Implement offline/failure handling.
25. Implement security.

26. Implement testing.
27. Implement demo simulation.
28. Run the complete end-to-end flow.
29. Red-team the system.
30. Fix the issues.
31. Polish the UI.
32. Verify the final demo.
33. Document everything.

At every stage ask:

“What happens if this fails?”

and:

“What happens if the marine situation changes after ORCA’s original recommendation?”

The second question is the defining principle of ORCA.

Build the application around:

**PLAN → PREDICT → DECIDE → MONITOR → DETECT → ALERT → GUIDE →
RESPOND → RESCUE → RECOVER → RETURN**

Do not stop until the complete end-to-end application and its critical edge cases are implemented and tested.